Depth-four set-multilinear formulas for sparse permanent

October 29, 2026

For a fixed bipartite support graph, the sparse-permanent deformation gives more than an evaluation algorithm: it gives a small algebraic formula for the entire weighted perfect-matching polynomial. This note proves a depth-four, syntactically set-multilinear version and tracks its dependence on the maximum degree. The proof uses the deformation and interpolation framework of [Exp26]; the required algebraic identities and the degree-dependent estimates are established below.

1 Statement and formula model

Let $G \subseteq [n] \times [n]$ be a bipartite graph. Give each edge $(u, v) \in G$ its own variable x_{uv}, and define

$$A_G(x)_{uv} = \begin{cases} x_{uv}, & (u, v) \in G, \\ 0, & (u, v) \notin G. \end{cases} \qquad P_G(x) = \operatorname{per} A_G(x).$$

The variable groups are the rows:

$$\mathcal{X}_u = \{x_{uv} : (u, v) \in G\}, \qquad u \in [n].$$

If G has maximum degree at most Δ, then $|\mathcal{X}_u| \le \Delta$ and every column also has at most Δ incident edges. All logarithms below are base two.

Theorem 1.1 (Growing maximum degree)**.** Let $\Delta = \Delta(n) \ge 2$ be integer-valued, and put

$$\beta = \log(\Delta + 1) + \log\log n.$$

Suppose $\beta = o(\sqrt{\log n})$. For every bipartite graph G with n vertices on each side and maximum degree at most Δ, the polynomial P_G has a division-free, homogeneous, syntactically row-set-multilinear depth-four formula over $\mathbb{Q}$ of size

$$\boxed{2^{O\left(\frac{n\beta}{\sqrt{\log n}} \log \frac{\sqrt{\log n}}{\beta}\right)}.} \tag{1.1}$$

The constants in the bound are absolute. The formula can be constructed deterministically from G within the same asymptotic bound in bit operations, and all its rational constants have polynomial bit length. Its bottom multiplication gates have variable fan-in

$$O\left(\frac{n}{\Delta(\log n)^{10}}\right). \tag{1.2}$$

All assertions are for sufficiently large n under the stated growth condition.

The condition is equivalent to $\Delta = 2^{o(\sqrt{\log n})}$. Since $a \log(1/a) = o(1)$ when $a \to 0$, the bound in (1.1) is subexponential in n.

Corollary 1.2. For every fixed C, graphs of maximum degree at most $(\log n)^C$ have formulas of size

$$2^{O_C\left(n(\log\log n)^2/\sqrt{\log n}\right)}.$$

For every fixed $0 < \alpha < 1/2$, maximum degree at most $2^{(\log n)^\alpha}$ gives size

$$2^{O_\alpha\left(n \log\log n/(\log n)^{1/2-\alpha}\right)}.$$

1.1 Formula model

Definition 1.3 (Row-set-multilinearity)**.** A polynomial is *set-multilinear on a row set I* if each of its monomials contains exactly one variable from each $\mathcal{X}_u$, $u \in I$, and no other variables. A formula is *syntactically row-set-multilinear* if its gates can be labeled by row sets so that a variable from $\mathcal{X}_u$ has label $\{u\}$, constants have label $\emptyset$, sums have children with the same label, and products have children with pairwise disjoint labels, whose union is the parent's label. Zero expressions may carry any row-set label.

We use unbounded-fan-in arithmetic formulas over $\mathbb{Q}$. A depth-four formula has shape $\Sigma\Pi\Sigma\Pi$, with constants and variables at its leaves. Size counts gates and leaf occurrences; there is no sharing of subformulas. Rational weights on sums can equivalently be absorbed into constant leaves without adding a layer.

Concretely, the formulas constructed here have the form

$$\sum_{\nu=1}^{R} \prod_{B \in \mathcal{P}_\nu} \left(\sum_{\sigma \in \prod_{u \in B} N_G(u)} c_{\nu,B,\sigma} \prod_{u \in B} x_{u,\sigma(u)} \right), \tag{1.3}$$

where $\mathcal{P}_\nu$ is a partition of $[n]$ into nonempty row sets. Zero coefficients are omitted. The partitions may differ between top-level summands; no common partition is asserted. Formula (1.3) is homogeneous and syntactically row-set-multilinear. It need not be set-multilinear in the *column* groups.

The formula depends on the support graph G, but works for all edge weights, including zero weights. It is not a single formula in n^2 variables promised that its input has sparse support.

Proof structure. First we turn a block-diagonal matrix plus a row-linear low-rank term into a depth-four formula. Next we show that Taylor coefficients and Hermite interpolation preserve this structure. We then construct the requisite polynomial matrix and choose its parameters. The key invariant is that an entry in row u is always linear in $\mathcal{X}_u$; no division by an expression in the edge variables is used.

2 Structured permanents as depth-four formulas

Call a matrix *row-linear* if each entry in row u is a homogeneous linear form in $\mathcal{X}_u$. Row-linear matrices need not have the original sparse support: their rows may become dense, but still involve at most Δ variables apiece. Empty products and empty permanents are 1.

Lemma 2.1 (Small blocks and a row-linear rank factorization)**.** Let $C(x)$ be a square matrix with row set $I \subseteq [n]$ and column set $J \subseteq [n]$. Suppose

$$C(x) = B(x) + U(x)V^{\mathsf{T}},$$

where B, U are row-linear, V is a constant rational matrix with r columns, and B has corresponding diagonal blocks $I_b \times J_b$ with $|I_b|, |J_b| \le s$. Corresponding row and column groups need not have equal sizes.

Then $\operatorname{per} C(x)$ has a homogeneous, syntactically row-set-multilinear $\Sigma\Pi\Sigma\Pi$ formula of size

$$\operatorname{poly}(n, r)\ (n+1)^{2r}\ 2^{O(s \log(\Delta+1))}. \tag{2.1}$$

Each upper product combines polynomials on the nonempty row sets I_b; the bottom multiplication gates have variable fan-in at most s. Given the matrices, the formula can be constructed within the bound (2.1), up to polynomial bit-arithmetic overhead.

Proof. Introduce auxiliary tuples z, w of length r, and put

$$u_u(x; z) = \sum_{a=1}^{r} U_{ua}(x) z_a, \qquad v_\nu(w) = \sum_{a=1}^{r} V_{\nu a} w_a.$$

For one block define

$$F_b(x; z, w) = \sum_{\substack{S \subseteq I_b,\ T \subseteq J_b \\ |S|=|T|}} \operatorname{per} B(x)[S, T] \prod_{u \in I_b \setminus S} u_u(x; z) \prod_{v \in J_b \setminus T} v_v(w). \tag{2.2}$$

Let $F = \prod_b F_b$. For multi-indices $\mu, \nu \in \mathbb{Z}^r_{\geq 0}$, define the $\mathbb{Q}[x]$-linear functional

$$\mathcal{L}_r(z^\mu w^\nu) = \begin{cases} \mu! := \prod_a \mu_a!, & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases} \tag{2.3}$$

Then

$$\operatorname{per} C(x) = \mathcal{L}_r(F). \tag{2.4}$$

To verify this, choose the edges supplied by B within each block. Color each remaining endpoint by one of the r outer-product types in UV^T. Endpoints of a given color must occur equally often on the two sides, and μ_a endpoints of color a can be paired in $\mu_a!$ ways. This is the color-count identity behind the fixed-rank permanent algorithm [Bar96]. It is an identity in $\mathbb{Q}[x]$, so signs of coefficients cause no difficulty.

Replace the functional by numerical evaluations. Every individual auxiliary variable in F has degree at most n. For $a \in \{0, \ldots, n\}$, let

$$\Lambda_a(t) = \prod_{\substack{0 \leq c \leq n \\ c \neq a}} \frac{t - c}{a - c}, \qquad \kappa_{ab} = \sum_{k=0}^{n} k!\, [t^k]\Lambda_a(t)\, [t^k]\Lambda_b(t).$$

Tensor-product Lagrange interpolation, followed by (2.3), gives

$$\operatorname{per} C(x) = \sum_{a,b \in \{0,\ldots,n\}^r} \left(\prod_{c=1}^{r} \kappa_{a_c b_c} \right) \prod_j F_j(x; a, b). \tag{2.5}$$

Indeed, the functional on $\Lambda_{a_c}(z_c)\Lambda_{b_c}(w_c)$ is $\kappa_{a_c b_c}$, and the pairing factors over the r coordinates. There are $(n+1)^{2r}$ evaluations. For $r = 0$, there is a single evaluation and its weight is 1.

Expand only the local factors. Fix numerical a, b. In every summand of (2.2), a row of I_j contributes once: either inside the small permanent or through u_u. Thus $F_j(x; a, b)$ is set-multilinear on I_j. It has at most

$$\prod_{u \in I_j} |\mathcal{X}_u| \leq \Delta^{|I_j|} \leq \Delta^s$$

possible monomials. Write it explicitly as a sum of those monomials.

Equation (2.5) now has four layers: a sum over the evaluation grid, a product over blocks, a sum of local monomials, and the product of variables in each monomial. Distinct blocks have disjoint row sets. Blocks with no rows contribute scalars, which can be absorbed into another factor. The scalar weight of a grid point can be absorbed in the same way. This proves the syntactic properties and the size bound.

For constructibility, enumerate the at most 2^{2s} pairs S, T in each block and use Ryser's formula for their permanents. Carry out the resulting polynomial arithmetic in the explicit row-set-multilinear monomial basis, which has at most Δ^s elements. This takes $\operatorname{poly}(n, r)2^{O(s \log(\Delta+1))}$ operations per grid point. It produces the coefficients of the bottom sums directly; no unfolding of a dynamic-programming circuit is needed. □

The hypothesis is an explicit factorization with *constant* V and row-linear U. In the application this factorization comes from a constant intermediate error matrix, rather than Gaussian elimination over $\mathbb{Q}(x)$.

3 Taylor coefficients and interpolation preserve depth four

Let $A(t,x)$ be an $n \times n$ matrix whose entries are polynomials in t of degree at most d and homogeneous linear forms in their respective row-variable groups. Put

$$P(t,x) = \operatorname{per} A(t,x), \qquad \deg_t P \le nd.$$

Proposition 3.1 (Algebraic recovery from structured points)**.** Suppose that for $i = 1, \dots, \ell$ we have

$$A(i,x) = B_i(x) + U_i(x)V_i^{\mathsf{T}},$$

with the structure of Lemma 2.1, uniformly bounded by block size s and r columns in the rank factors. Let

$$h = \left\lfloor \frac{nd}{\ell} \right\rfloor + 1 \le n/2.$$

Then $P(0,x)$ has a homogeneous, syntactically row-set-multilinear depth-four formula of size

$$\operatorname{poly}(n,\ell,d,r)\,(n+1)^{2r}\,2^{O(s\log(\Delta+1))}\sum_{a=0}^{h-1} 2^a \binom{n}{a}^2 \tag{3.1}$$

$$\le \operatorname{poly}(n,\ell,d,r)\,2^{O(s\log(\Delta+1)+r\log n+h\log(n/h))}. \tag{3.2}$$

Each upper product uses the remaining blocks of one B_i, together with fewer than h additional linear factors on distinct rows. The bottom variable fan-in is at most $\max\{s,1\}$.

Proof. Fix i and write

$$A_i(x) = A(i,x), \qquad D_i(z,x) = A(i+z,x) - A_i(x).$$

Every entry of D_i is divisible by z. The permanent addition formula yields

$$P(i+z,x) = \sum_{\substack{S,T\subseteq[n]\\ |S|=|T|}} \operatorname{per} A_i(x)[S^c,T^c]\ \operatorname{per} D_i(z,x)[S,T].$$

If $|S| = |T| = a$, the second permanent is divisible by z^a. Therefore, for $0 \le j < h$,

$$[z^j]P(i+z,x) = \sum_{a=0}^{j} \sum_{\substack{S,T\subseteq[n]\\ |S|=|T|=a}} \operatorname{per} A_i(x)[S^c,T^c]\ [z^j]\operatorname{per} D_i(z,x)[S,T]. \tag{3.3}$$

The first factor uses exactly the rows in S^c; the second uses exactly those in S.

The first factor has the formula from Lemma 2.1: deleting rows and columns preserves the block partition and restricts U_i, V_i. Its local row sets are the original row blocks with S removed.

For the second factor, the polynomial in z has degree at most ad. Consequently there are rational constants $\theta_{a,j,v}$ such that

$$[z^j]\operatorname{per} D_i(z,x)[S,T] = \sum_{v=0}^{ad} \theta_{a,j,v}\ \operatorname{per} D_i(v,x)[S,T]. \tag{3.4}$$

These are the coefficients of the univariate Lagrange basis on $\{0,\dots,ad\}$. Coefficients with $j > ad$ are zero. The case $a = 0$ is the constant polynomial 1.

For each numerical v, Ryser's formula [Rys63] gives

$$\operatorname{per} D_i(v,x)[S,T] = \sum_{W\subseteq T} (-1)^{a-|W|} \prod_{u\in S} \left(\sum_{c\in W} D_i(v,x)_{uc} \right). \tag{3.5}$$

Each parenthesized expression is a homogeneous linear form in $\mathcal{X}_u$. Keep these linear forms as factors; do *not* expand their product into monomials.

Substitute (2.5), (3.4), and (3.5) into (3.3), and distribute only the outer sums. Each resulting product consists of local row-block polynomials on S^c and singleton-row linear forms on S. Their row sets partition $[n]$. The local polynomials are bottom sums of monomials, and the linear forms are bottom sums of degree-one monomials. Thus this is still a syntactically row-set-multilinear $\Sigma\Pi\Sigma\Pi$ formula. In particular, there is no Δ^a cost from expanding the product over the affected rows.

There are $\binom{n}{a}^2$ choices of S, T, 2^a choices of W, and polynomially many choices of v, i, j. Lemma 2.1 supplies the other factors in (3.1). For $h \le n/2$,

$$\sum_{a<h} 2^a \binom{n}{a}^2 \le h\, 2^h \binom{n}{h}^2 = 2^{O(h\log(n/h))}.$$

Finally, $\ell h > nd$. Hermite interpolation gives rational constants $\lambda_{i,j}$, independent of x, for which

$$P(0,x) = \sum_{i=1}^{\ell} \sum_{j=0}^{h-1} \lambda_{i,j}[z^j] P(i+z,x). \tag{3.6}$$

This identity holds coefficientwise in $\mathbb{Q}[x]$. Its sum merges into the top layer, proving the proposition. □

The depth-four conclusion comes from these explicit identities, not merely from the existence of a small arithmetic circuit or a polynomial-space evaluation algorithm.

4 Factoring a sparse support graph

The next construction produces the row-linear polynomial matrix needed in Proposition 3.1. Its combinatorial choices depend only on G.

4.1 Balanced word labels

Choose integers $q \ge 2$ and $\ell \ge 1$, and write $Q = q^\ell$.

Lemma 4.1 (Balanced labeling)**.** If

$$4\Delta\sqrt{n} \le Q \le n, \qquad q \ge 4\Delta\ell, \tag{4.1}$$

then the rows and columns can be labeled by words $\alpha(u), \gamma(v) \in [q]^\ell$ such that:

(i) each word occurs at most $D = \lceil 2n/Q \rceil$ times on each side;

(ii) distinct edges have distinct ordered word pairs $(\alpha(u), \gamma(v))$;

(iii) for every edge (u,v) and every $i \in [\ell]$, $\alpha(u)_i \ne \gamma(v)_i$.

The labels can be found deterministically in polynomial time.

Proof. Stop using a word on a side after D uses. Before the final vertex is assigned, fewer than $Q/2$ words on that side are full. Label the rows first, requiring rows with a common neighbor to have distinct labels. A row conflicts with at most $\Delta(\Delta-1)$ previous rows.

Now label columns. A column has at most Δ neighbors, carrying distinct row labels. Each neighboring row label occurs on at most D rows, hence has already been paired with at most ΔD column labels. Avoiding repeated ordered pairs forbids at most $\Delta^2 D$ choices. Enforcing coordinatewise disagreement forbids at most $\Delta \ell Q/q \le Q/4$ further choices.

The inequalities in (4.1) imply $Q \ge 16\Delta^2$ and $Q^2 \ge 16\Delta^2 n$. Therefore

$$\Delta^2 D \le \frac{2\Delta^2 n}{Q} + \Delta^2 \le \frac{Q}{8} + \frac{Q}{16} < \frac{Q}{4}.$$

Together with fewer than $Q/2$ full labels, these exclusions leave a valid choice. They also justify the initial row-labeling step. Choosing the first valid word gives the claimed algorithm. □

For each coordinate i and symbol a, define

$$I_{i,a} = \{u : \alpha(u)_i = a\}, \qquad J_{i,a} = \{v : \gamma(v)_i = a\}.$$

There are Q/q words with a fixed symbol in one position, so

$$|I_{i,a}|, |J_{i,a}| \le \frac{Q}{q}\left\lceil \frac{2n}{Q} \right\rceil \le \frac{3n}{q}. \tag{4.2}$$

4.2 Ordered word pairs and the middle matrix

Let

$$\Omega = \{(a,b) \in [q]^\ell \times [q]^\ell : a_i \ne b_i \text{ for every } i\}, \qquad M = (q(q-1))^\ell.$$

For each edge (u,v), set

$$L(x)[u, (\alpha(u), \gamma(v))] = x_{uv}, \qquad R[(\alpha(u), \gamma(v)), v] = 1,$$

with all other entries zero. Uniqueness of ordered label pairs gives $A_G(x) = L(x)R$.

In each coordinate, an ordered pair of distinct symbols consists of an unordered pair and an orientation bit. Consequently

$$\Omega \cong C \times \{0,1\}^\ell, \qquad C = \left(\binom{[q]}{2}\right)^\ell, \qquad M = |C|2^\ell.$$

Use increasing order for orientation bit 0 and decreasing order for bit 1. Put

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathcal{J} = I_C \otimes X^{\otimes \ell}, \qquad R' = \mathcal{J}R.$$

The matrix $\mathcal{J}$ reverses the two words of every ordered pair. Since $\mathcal{J}^2 = I$,

$$A_G(x) = L(x)R = L(x)\mathcal{J}R'. \tag{4.3}$$

The factors R' and $\mathcal{J}$ are constant; $L(x)$ is row-linear.

A matrix K on $\{0,1\}^\ell$ is *i-nice* if it is $I_2 \otimes B$ after moving tensor factor i first. In particular, $K[b,c] = 0$ whenever $b_i \ne c_i$.

Lemma 4.2 (One unchanged coordinate gives blocks)**.** For every i-nice K, the row-linear matrix

$$L(x)(I_C \otimes K)R'$$

has corresponding diagonal blocks $I_{i,a} \times J_{i,a}$, $a \in [q]$. Each has at most $3n/q$ rows and columns.

Proof. A contribution to entry (u, v) uses an edge (u, v') in L and an edge (u', v) in R'. Their intermediate labels are

$$(\alpha(u), \gamma(v')) \quad \text{and} \quad (\gamma(v), \alpha(u')).$$

The middle matrix stays inside one cube, so the unordered symbol pairs at position i agree. It preserves orientation bit i, so the ordered pairs agree. Their first symbols give $\alpha(u)_i = \gamma(v)_i$. This is exactly the asserted block support; sizes follow from (4.2). □

5 A low-degree matrix polynomial with a tunable rank error

We need to vary the exceptional fraction with Δ, rather than fix it to a power of ℓ.

Lemma 5.1 (Parity agreement)**.** For $0 < \varepsilon < 1/2$, set

$$k_0 = \left\lceil \log \frac{2}{\varepsilon} \right\rceil, \qquad w = \left\lceil \sqrt{\ell k_0} \right\rceil, \qquad d = 2w. \tag{5.1}$$

There is a rational polynomial p of degree at most d such that

$$\Pr_{x \in \{-1,1\}^\ell} \left[p\left(\sum_j x_j \right) \neq \prod_j x_j \right] \leq \varepsilon. \tag{5.2}$$

Proof. If x has k negative coordinates, then its sum is $\ell - 2k$ and its parity is $(-1)^k$. Interpolate $p(\ell - 2k) = (-1)^k$ for all integer $k \in [0, \ell]$ with $|k - \ell/2| \leq w$. There are at most $2w + 1$ interpolation points. For $k \sim \mathrm{Bin}(\ell, 1/2)$, the exceptional probability is at most

$$2\exp(-2w^2/\ell) \leq 2^{1-k_0} \leq \varepsilon.$$

The interpolation points are distinct integers, so the coefficients are rational. □

This is exact agreement outside an exceptional set; the values on that set are not required to be bounded.

For $j \in [\ell]$, let X_j be X on tensor factor j and the identity on the other factors. Define

$$\begin{aligned} \widetilde{J}(t) &= p\left(\sum_{j=1}^{\ell} (1 - t/j) X_j \right), \\ E_0 &= X^{\otimes \ell} - p\left(\sum_{j=1}^{\ell} X_j \right), \qquad J(t) = \widetilde{J}(t) + E_0. \end{aligned} \tag{5.3}$$

Lemma 5.2 (Local deformation)**.** The matrix polynomial $J(t)$ has degree at most d, satisfies $J(0) = X^{\otimes \ell}$, and for every $i \in [\ell]$ has the form

$$J(i) = K_i + E_0, \qquad K_i \text{ is } i\text{-nice}, \qquad \mathrm{rank}(E_0) \leq \varepsilon 2^\ell.$$

All matrices and polynomial coefficients are rational and explicitly constructible.

Proof. The degree and the identity at zero are immediate from (5.3). At $t = i$, the coefficient of X_i vanishes. After moving factor i first, the matrix inside p has the form $I_2 \otimes S_i$, so

$$K_i = \widetilde{J}(i) = I_2 \otimes p(S_i).$$

For the rank bound, let

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The identities $H^2 = I_2$ and $HXH = Z$ give

$$H^{\otimes \ell} E_0 H^{\otimes \ell} = Z^{\otimes \ell} - p\left(\sum_j Z_j\right).$$

This is a diagonal matrix with entries $\prod_j x_j - p(\sum_j x_j)$, one for each $x \in \{-1, 1\}^\ell$. At most $\varepsilon 2^\ell$ entries are nonzero by (5.2). Conjugation preserves rank. Formula (5.3) itself uses only rational arithmetic; the square roots appear only in this proof. □

5.1 Lift the local deformation to the symbolic input

Set

$$A(t, x) = L(x)(I_C \otimes J(t))R'. \tag{5.4}$$

Then $A(0, x) = A_G(x)$, and every entry has degree at most d in t and is homogeneous linear in its row's variables.

The constant matrix $\mathcal{E} = I_C \otimes E_0$ has rank

$$r_0 \le |C|\varepsilon 2^\ell = \varepsilon M.$$

Choose a rational rank factorization $\mathcal{E} = F_0 G_0^{\mathsf{T}}$, and define

$$U(x) = L(x)F_0, \qquad V = R'^{\mathsf{T}} G_0.$$

Thus U is row-linear and V is constant. At each special point,

$$A(i, x) = \underbrace{L(x)(I_C \otimes K_i)R'}_{B_i(x)} + U(x)V^{\mathsf{T}}. \tag{5.5}$$

Lemma 4.2 gives block size at most $\lceil 3n/q \rceil$. The same rank factors work at every point. Crucially, the only rank factorization performed is on the *constant rational matrix* $\mathcal{E}$.

Theorem 5.3 (Parameter form)**.** Suppose q, ℓ satisfy (4.1), choose $0 < \varepsilon < 1/2$, and define d by (5.1). Put

$$M = (q(q-1))^\ell, \qquad s = \left\lceil \frac{3n}{q} \right\rceil, \qquad h = \left\lfloor \frac{nd}{\ell} \right\rfloor + 1.$$

If $h \le n/2$, then P_G has a homogeneous, syntactically row-set-multilinear depth-four formula of size

$$\boxed{\operatorname{poly}(n)\, 2^{O(s \log(\Delta+1) + \varepsilon M \log n + h \log(n/h))}.} \tag{5.6}$$

Its bottom variable fan-in is at most s. Each upper product contains at most q block factors and fewer than h additional linear factors. All coefficients are rational.

Proof. Apply Proposition 3.1 to (5.4), using the decompositions (5.5) and $r_0 \le \varepsilon M$. All auxiliary dimensions are polynomial in n: $M \le Q^2 \le n^2$, $\ell \le \log n$, and $h \le n/2$ forces $d < \ell/2$. This absorbs the proposition's polynomial prefactor. □

6 Choosing parameters for maximum degree Δ

We prove Theorem 1.1. Write $\lambda = \lceil \log n \rceil$. Then

$$\beta = \Theta\big(\log(\Delta + 1) + \log \lambda\big) = o(\sqrt{\lambda}).$$

Choose

$$q = \Delta\lambda^{10}, \qquad \ell = \left\lceil \log_q(16\Delta\sqrt{n}) \right\rceil, \qquad \varepsilon = \Delta^{-4}\lambda^{-100}, \tag{6.1}$$

and let d be as in (5.1).

The labeling conditions. The choice of ℓ gives

$$16\Delta\sqrt{n} \le Q = q^{\ell} < 16\Delta q\sqrt{n} = 16\Delta^2\lambda^{10}\sqrt{n}.$$

Since $\log \Delta = o(\sqrt{\lambda})$, the last expression is less than n. Also $\ell = O(\lambda)$ and $q \ge 4\Delta\ell$ for sufficiently large n. Thus (4.1) holds. Moreover,

$$\ell = \Theta(\lambda/\beta), \qquad M \le Q^2 = O(n\Delta^4\lambda^{20}). \tag{6.2}$$

Block size and rank error. Equations (6.1)–(6.2) give

$$s = O\Big(\frac{n}{\Delta\lambda^{10}}\Big), \qquad \varepsilon M = O(n/\lambda^{80}). \tag{6.3}$$

In particular,

$$s \log(\Delta + 1) = O(n/\lambda^{10}), \qquad \varepsilon M \log n = O(n/\lambda^{79}).$$

The factor Δ^{-4} in the exceptional fraction compensates for the degree dependence of the intermediate dimension.

Degree and number of Taylor coefficients. Since

$$\log(1/\varepsilon) = 4\log\Delta + 100\log\lambda = \Theta(\beta),$$

we have

$$d = \Theta\Big(\sqrt{\ell\beta}\Big) = \Theta(\sqrt{\lambda}), \qquad \frac{d}{\ell} = \Theta\left(\frac{\beta}{\sqrt{\lambda}}\right) = o(1).$$

Therefore

$$h = \left\lfloor \frac{nd}{\ell} \right\rfloor + 1 = \Theta\left(\frac{n\beta}{\sqrt{\lambda}}\right) \le n/2 \tag{6.4}$$

for sufficiently large n. Its contribution to the exponent is

$$h \log(n/h) = O\left(\frac{n\beta}{\sqrt{\lambda}} \log \frac{\sqrt{\lambda}}{\beta}\right). \tag{6.5}$$

Because $\beta = \Omega(\log\lambda)$ and $\beta/\sqrt{\lambda} = o(1)$, this bound also absorbs n/λ^{10}, n/λ^{79}, and polynomial prefactors. Substituting into Theorem 5.3 proves (1.1); (6.3) proves (1.2). The corollary follows by inserting the indicated degree bounds.

6.1 Uniform construction and exact constants

The label assignment is a deterministic polynomial-time computation on G. The local parity polynomial is obtained by interpolation at at most $\ell + 1$ integers of magnitude at most ℓ. Its coefficients have polynomial bit length. The substitutions in (5.3) use rational numbers with denominators at most ℓ, and all matrix dimensions and polynomial degrees are polynomial in n.

The rank factorization of $\mathcal{E}$ is computed over $\mathbb{Q}$, before introducing the edge variables. Standard determinant bounds for rational linear algebra give polynomial-bit factors. Thus all row-linear coefficients in B_i, U, V have polynomial bit length. The Lagrange and Hermite constants also come from rational systems of polynomial dimension and polynomial-bit entries, so their bit lengths are polynomial.

To construct each formula, use the explicit grid, subset, and interpolation sums above. A local coefficient is a sum of products of at most polynomially many polynomial-bit rationals; there are at most exponentially many such products with a polynomial-size exponent. A common denominator of polynomial bit length can be chosen throughout each coefficient computation. Hence the emitted coefficients have polynomial bit length. The symbolic local expansions have cost $2^{O(s \log(\Delta+1))}$, and the other enumerations have exactly the costs in (3.1). The resulting construction time is bounded by (1.1), with a change in its absolute constant.

No arithmetic operation divides by an edge variable or a polynomial in edge variables. All divisions have already been incorporated into rational constants. The output is therefore a division-free formula, not a rational-function circuit.

7 What the formula structure asserts

The variable partition into the original row groups is fixed throughout. At every product gate, its children use disjoint sets of those groups. At every sum gate, all children use the same set. This proves syntactic set-multilinearity, not just multilinearity of the final answer, and also proves homogeneity at every gate.

The coarser partition into row blocks is allowed to vary with the interpolation point and the deleted row set. Thus the theorem does not impose a common block partition on all top product gates. Each individual summand nevertheless has the explicit form (1.3).

The formula uses signed rational coefficients. Intermediate monomials may select the same column more than once; cancellation enforces the perfect-matching condition. The result is neither a monotone formula nor a claim of simultaneous syntactic set-multilinearity in both the row and column partitions.

The three structural steps are exact: numerical pairing converts low rank into a top-level sum, small row blocks give sparse local polynomials, and Taylor coefficient extraction introduces only singleton-row linear factors. These are what make the fourth layer sufficient.

References

[Bar96] Alexander I. Barvinok. Two Algorithmic Results for the Traveling Salesman Problem. *Mathematics of Operations Research* 21(1):65–84, 1996. doi:10.1287/moor.21.1.65.

[Exp26] *Some exposition of "The Exponential Time Hypothesis Is False."* Supplied exposition dated October 29, 2026. Sections 2–6 and Appendix A.

[Rys63] Herbert John Ryser. *Combinatorial Mathematics*. Carus Mathematical Monographs, volume 14. Mathematical Association of America, 1963.